

Lab - 7 : Modelling strategic conflict between nations

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 CS302, Modelling and Simulation*

In this lab we model and plot the war-waging potential of two rival countries. This depends on various factors like war readiness of the rival country, grievance it feels towards the rival country and armament cost of it's own. These parameters affect the war-waging potential of countries.

I. EQUATIONS

Strategic conflict between two nations is captured by the coupled equations

$$\dot{x} = ky + g - \alpha x \quad (1)$$

$$\dot{y} = lx + h - \beta y \quad (2)$$

For Mutual disarmament without grievance,

$$g = h = 0$$

$$\dot{x} = ky - \alpha x \quad (3)$$

$$\dot{y} = lx - \beta y \quad (4)$$

For Mutual disarmament with grievance,

$$g, h \neq 0$$

$$y(0) = x(0) = 0 \quad (5)$$

$$\dot{x} = g \quad (6)$$

$$\dot{y} = h \quad (6)$$

For Unilateral disarmament,

$$y(0) = 0, x(0) \neq 0 \quad (7)$$

$$\dot{y} = lx + h \quad (7)$$

For Arms race

$$g = h = 0$$

$$\alpha = \beta = 0 \quad (1)$$

$$\dot{x} = ky \quad (2)$$

$$\dot{y} = lx \quad (2)$$

II. RESULTS

A. Mutual disarmament without grievance

Fig. 1 shows the Disarmament of countries x and y, without grievance with respect to time.

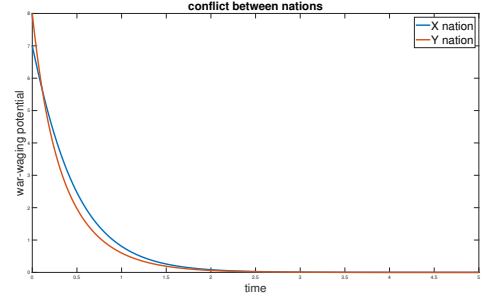


FIG. 1: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units.

Fig. 2 shows the Disarmament of countries x and y, without grievance with respect to time, taking logarithmic scale on y-axis.

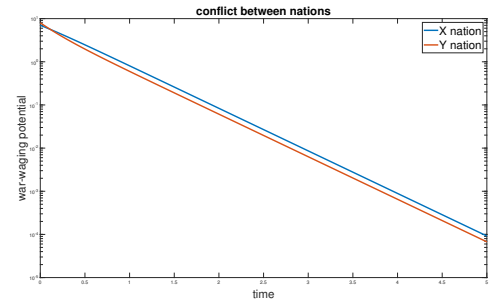


FIG. 2: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units

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Fig. 3 shows the Disarmament of countries y vs x, without grievance.

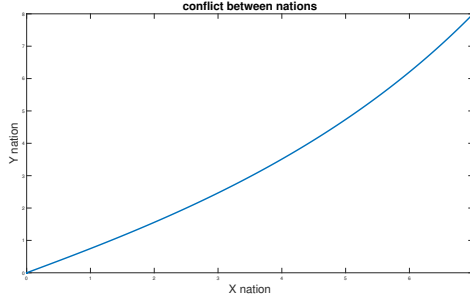


FIG. 3: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units.

B. Mutual disarmament with grievance

Fig. 4 shows the Disarmament of countries x and y, with grievance with respect to time.

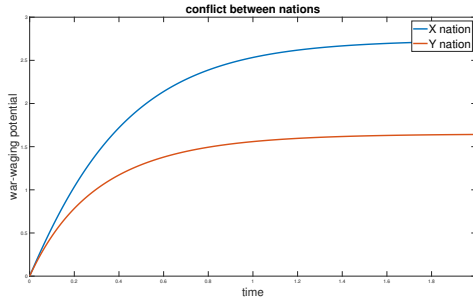


FIG. 4: Here, $x(0) = 0$, $y(0) = 0$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 6$, $h = 5.5$ and $\Delta t = 0.01$ units.

Fig. 5 shows the Disarmament of countries y vs x, with grievance.

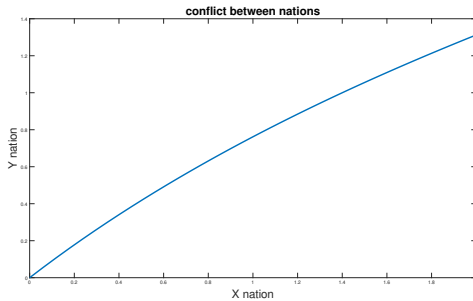


FIG. 5: Here, $x(0) = 0$, $y(0) = 0$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 6$, $h = 5.5$ and $\Delta t = 0.01$ units.

C. Unilateral disarmament

Fig. 6 shows the Unilateral Disarmament of country y with respect to time.

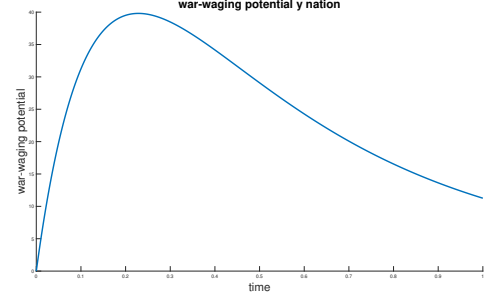


FIG. 6: Here, $x(0) = 7$, $y(0) = 0$, $\alpha = 4$, $\beta = 5$, $k = 3$, $l = 2$, $g = 6$, $h = 5.5$ and $\Delta t = 0.01$ units.

D. Arms race

Fig. 7 shows the Arms Race of countries x and y, with respect to time.

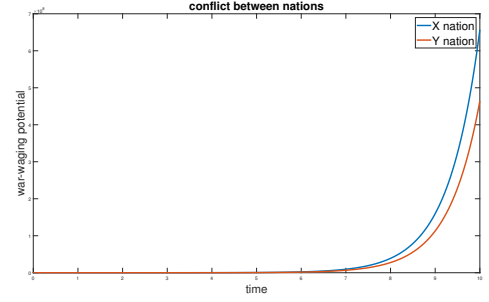


FIG. 7: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 0$, $\beta = 0$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units.

Fig. 8 shows the Arms Race of countries x and y, with respect to time, taking logarithmic scale on y-axis.

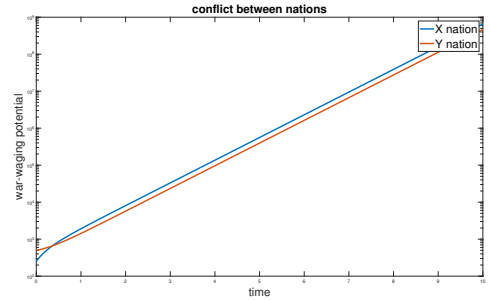


FIG. 8: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 0$, $\beta = 0$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units.

Fig. 9 shows the Arms Race of countries y vs x.

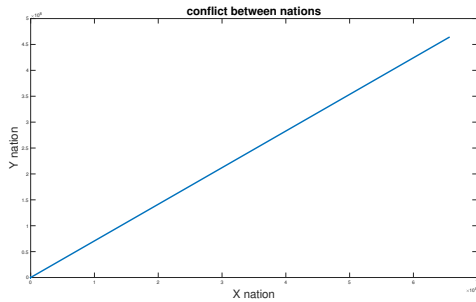


FIG. 9: Here, $x(0) = 7$, $y(0) = 8$, $\alpha = 0$, $\beta = 0$, $k = 3$, $l = 2$, $g = 0$, $h = 0$ and $\Delta t = 0.01$ units.

III. CONCLUSIONS

- **Effect of Grievance on War-Waging Potential:** We find that in the absence of any grievance, the war-waging potential of countries diminishes exponentially over time. However, the presence of grievance leads to a linear increase in war-waging potential.
- **Impact of War-Readiness Factor on War-Waging Potential:** Our observations reveal that the war-readiness factor of a rival country prompts an exponential escalation in war-waging potential, irrespective of any existing grievance between the nations.
- **Relation between Armament and Rival's Armament:** We note a linear correlation between a country's armament and that of its rival. As the armament of one country increases, so does the armament of its rival nation.